

Skill Arena - Phase 5 Master Overview

Technical Architecture, Platform Engineering & Infrastructure Blueprint

Phase 5 Purpose

Define how Skill Arena is engineered, deployed, scaled and operated at a technical level.

Primary Objective

Provide a complete technical blueprint for developers, architects and Codex to build the platform.

Architecture Philosophy

Microservice-driven, event-driven, cloud-native and server-authoritative.

Document 1

System Architecture, Backend Services & Microservices Design.

Document 2

API Architecture, Service Communication, Event Bus Design & Real-Time Messaging Framework.

Document 3

Game Engine Architecture, Matchmaking Engine, Challenge Generation & Gameplay Processing Framework.

Document 4

Database Architecture, Caching Strategy, Replay Storage & High Availability Framework.

Document 5

Cloud Infrastructure, Deployment Architecture, CI/CD Pipeline, DevOps & Global Scaling Framework.

Core Platform Services

User Service, Gameplay Service, Matchmaking Service, Wallet Service, Treasury Service, Tournament Service, Replay Service, Security Service and Compliance Service.

Communication Architecture

REST APIs, WebSockets, Event Bus Messaging and Service-to-Service Communication.

Gameplay Infrastructure

Procedural challenge generation, matchmaking, gameplay validation and replay systems.

Data Architecture

Domain-owned databases, caching layers, replay storage and analytics systems.

Scalability Model

Independent service scaling, global deployment support and high-throughput event processing.

Reliability Model

High availability, failover support, disaster recovery and operational resilience.

Observability Framework

Logging, tracing, monitoring, alerting and operational dashboards.

Security Integration

Authentication, authorization, encryption, service security and infrastructure protection.

Deployment Model

Automated CI/CD, containerized workloads, orchestration and infrastructure as code.

Expected Outcome

A production-ready technical architecture capable of supporting a global competitive gaming ecosystem.

Transition To Phase 6

Phase 6 defines Product Expansion, Multi-Game Framework, AI Systems, Analytics, Marketplace Features and Long-Term Platform Evolution.